

Alex Levesque

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EDUCATION

Queen’s University

Kingston, ON

BASc Applied Mathematics and Computer Engineering (GPA: 3.80/4.30)

Sep 2024 – Apr 2028

• Coursework: Algorithms, Engineering Design, Probability, Real Analysis, Microeconomics, Philosophy

AWARDS

GRAMMY Music Award, The Recording Academy

Feb 2024

• Received a GRAMMY Award for contributions to Killer Mike’s MICHAEL, 2024 Rap Album of the Year

• Produced for the Migos, Gunna, Peso Pluma, Roddy Ricch, Polo G, A Boogie, Lil Tjay, Skepta, Cam’ron

Canada’s Top Students, Scotiabank

Nov 2025

• Selected as 1 of 30 attendees (from 500+) for Scotiabank’s Canada’s Top Students conference and case competition

EXPERIENCE

Data Science Intern

May 2026 – Aug 2026

Scotiabank

Toronto, ON

• Designed and deployed an end-to-end, concurrent LLM parsing pipeline for unstructured audit files using Vertex AI and Openpyxl, cutting manual labor by 25%+ while holding processing time near-constant as file volume scaled

• Reduced \$70M+ in tech-risk loss exposure by building an HDBSCAN clustering pipeline, tuning it to minimize cluster overlap and cut LLM calls by 66.7%

• Developed an interactive Power BI expense analysis dashboard surfacing 4 KPIs across CAD spend and policy violations with dynamic 3-, 6-, and 12-month filtering, saving the expense team ~5 hours/week in manual review

Generative AI Engineering Intern

May 2025 – Aug 2025

Government of Canada

Ottawa, ON

• Designed and implemented an Azure-based GenAI workflow in PowerApps to transcribe, categorize, and organize refugee documents, achieving 93% accuracy and reducing manual processing time by over 50%

• Engineered an Azure RAG chatbot with an ETL pipeline, Cosmos DB, and vector storage to automate legal information retrieval, reducing average lookup time from hours to minutes

PROJECTS

For more projects and in-depth analyses regarding my projects, click here.

Markowitz Portfolio Optimizer (Convex Optimization)

• Built a Markowitz portfolio optimizer in Python with cvxpy, modeling risk minimization as a convex quadratic program that computes provably optimal allocations across 50+ assets in under 2 ms per solve

• Engineered a data pipeline pulling 5 years of market data via yfinance to estimate returns/covariance and trace a 50-portfolio efficient frontier in <0.1 s, validated by 14 automated tests across 4 modules

Adaptive Reinforcement Learning Ensemble Strategy, Queen’s AI Club (QMIND)

• Built and trained an ensemble of PPO, A2C, and TD3 reinforcement learning agents for daily portfolio allocation across 9 ETFs, using Stable-Baselines3 within a custom Gymnasium trading environment

• Validated backtests with 10,000-sample bootstrap confidence intervals and 10 bps transaction-cost modeling, confirming an annualized Sharpe of 1.14 at roughly half the drawdown of a passive S&P 500 benchmark

Autonomous Multi-Agent Robotic Firefighting, Second Year School Project

• Developed and optimized Lloyd’s algorithm in MATLAB for adaptive k-means clustering of dynamic fire hotspot data, reducing cluster convergence time by 30% and tightening robot-to-hotspot coverage

• Built the multi-agent coordination and simulation component in MATLAB, enabling 10 robots to cover 100% of hotspots in under 100 iterations

SKILLS & INTERESTS

Skills: Python, Excel, SQL, PowerBI, Microsoft Azure, Google Vertex AI, Cosmos DB, Copilot Studio

ML/AI: PyTorch, Scikit-learn, HDBSCAN, CVXPY, Pandas, NumPy, Jupyter Notebook, OpenPyXL

Interests: Native in English and French, Karate Black Belt, Cycling, Non-Fiction, Psychology